Device Network Topology Troubleshooting

Tags	collect_cdp.py, store_lldp_neighbors, IPAddressError: Address passed was not valid, collect_cdp_neighbors, store_lldp_neighbors, did_from_chassis, charmap, cdp_collect.py, crunch
Team Owner	Proc and Roll
Status	ACTIVE
Last Modified	Apr 4, 2024

This doc is targeted for the physical device topology, not the component devices

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Introduction

Here is another article on topology troubleshooting: Topology Troubleshooting Guide

Device topology collections use separate collection process for each type:

Collection Types	Process Name	Process File
CDP	Data Collection: CDP Collection	cdp_collect.py
LLDP	Data Collection: LLDP Collection	lldp_collect.py
L2 Topology	Data Collection: Topology	topology_collect.py
L3 Topology	Data Collection: L3 Topology Collection	l3topology_collect.py

All collection types use same process to process raw data: Enterprise Database: Topology Crunch (topology_crunch.py)

Note that the topology crunch process crunches the data sequentially in the order of L2, CDP, LLDP, L3.

Diagnosis and Mitigation

Engineering Needs For Diagnosis

In order to begin investigation escalations involved topology, we need the following data:

SL1 Logs With Debug Enabled For Topology Processes

To enable debug for topology processes, please follow the SL1 documentation here and provide engineering with the silo.log file(s) from the time period when topology collection took place.

We will need debug enabled for whichever topology processes are related to the escalation. For example, if it's an L3 topology issue, enable debug on the "Data Collection: L3 Topology Collection" process. The list of these processes can be found in the table above.

Topology Crunch Logs

After topology collection has run, please manually run topology crunch and provide us with the resulting output.

Topology crunch can be run on the central database appliance with the following command:

```
sudo -u s-em7-core SILO_DEBUG=1 /opt/em7/backend/topology_crunch.py > topology_crunch.log
```

Database Dump

After topology crunch has run, please provide us of a dump of the database tables where topology relationship data is stored.

This can be done by running the following command on the central database appliance:

```
mysqldump -u root -p master_dev device_cdp_map device_l3_map device_lldp_chassis
device_lldp_map topo_adhoc_relationships topo_cdp_relationships topo_comp_depth
topo_comp_relationships topo_complink_relationships topo_l2_multi topo_l2_relationships
topo_l3_relationships topo_lldp_relationships topo_vm_relationships > topo_data.sql
```

Please enter your root password when prompted from the above command. Send us the topo_data.sql file. This dump only contains topology related data for devices.

Optimization

This is an article that discusses optimization strategies for topology crunch: Optimization of Layer 2 Topology Crunch

Topology Crunch sigterms

We have seen customer systems with huge number of devices / interfaces generate large data of L2 topology, which blocks the topology crunch process to finish.

There are two crunch algorithms that user can choose, the new and legacy one, by toggling the config using SQL query below.

The new algorithm uses the recursive way to build the topology and could run into the "maximum recursion depth exceeded" issue when the topology tree is deep. In that case, you should see the exception or error message in the silo.log if you turn on the process debug mode. The **workaround** would be try to switch to the legacy crunch algorithm by running below query on CDB or AIO.

```
update master.system_custom_config set field_value = 1 WHERE field = 'use_legacy_l2_crunch'
```

In case switching back to legacy crunch algorithm still run into sigterm issue, it might be the process memory r_limit (by default 2GB) is not enough to load the large size of the data. Bump that up to maybe 4GB should handle most of the cases. If this still does not resolve the issue, please escalate to Engineering for more in-depth investigation.

To add a process memory limit override, update the /etc/silo.conf file and add a section like below format:

83 is the process id of the topology crunch backend process

```
[PROC_VIRTUAL_MEM_LIMIT]
83 = 2147483648
```

If customer runs into the sigterm issue before 8.12.2.4 or 8.14.0, they could try to upgrade to that version to consume the performance optimization change. If that's not an option:

Workaround: Ensure you actually need L2 topology data - if not, disable the Topology Collection process (despite the name, it's limited to just L2, not all topology).

And truncate the master_dev.device_interfaces_map table for L2 raw records. That table will always be repopulated in L2 topology collection.
Doing that will remove existing L2 raw data, and prevent em7 to collect L2 raw data and the crunch process will not do any L2 crunch.
The L2 relationship data (result from crunching L2 raw data) will then not get updated.
If they do need the L2 relationship, they could upgrade to 8.12.2.4+ or 8.14.0, we have fix (EM-32664) in those releases that help them not have crunch sigterm.

Topology Collection sigterms

If you see sigterms on topology collection process for certain topology type, it could be:

1. Too many devices/interfaces are to be collected
2. Slow SNMP response from certain devices

Enable the debug mode for the affected topology collection type and manually kick off the collection process at the Collector, and capture the complete debug log for the process run period.

Missing topology relationship:

If you can see some (or none) topology relationship for a certain type, but not seeing all relationship links you expected, please remember (using CDP as an example):

1. CDP must be turned on in SL1 settings > Behavior page
2. Your devices need to support CDP
3. CDP configuration must be activated on interfaces
4. The devices must be discovered by SL1 and interfaces must be inventoried

Check raw topology data tables

Run the collection process for each topology type will generate data in raw data tables.

Run the topology crunch process will generate data in topology relationship data tables (based on raw data).

If there's no relationship data for a certain topology type, please first check if there's raw topology data in corresponding tables. If raw data is absent, the issue may be in corresponding topology collection process. Otherwise, the issue lies in the Topology Crunch process.

	CDP	LLDP	L2	L3
raw data tables	master_dev.device_cdp_map	master_dev.device_lldp_chassis master_dev.device_lldp_map	master_dev.device_interfaces_map	master_dev.device_l3_map
processed data tables	master_dev.topo_cdp_relationships	master_dev.topo_lldp_relationships	master_dev.topo_l2_relationships master_dev.topo_l2_multi	master_dev.topo_l3_relationships

Collect information

Run the backend process manually in debug mode and get debug log like below.

For example, to kick off the CDP topology collection process at the Collector's terminal:

```
sudo -s (sudo to root first!)
sudo -u s-em7-core SILO_DEBUG=1 /opt/em7/backend/cdp_collect.py & tail -F /data/logs/silo.log
```

Also get related SQL dump of DB tables

How to enable storage log

To capture storage logs from CDB, add following block in silo.conf on the CDB.

```
[DATAPULL]
log_storage=1
```

Restart em7_hfpulld, em7_mfpulld, and em7_lfpulld on the CDB via systemctl

The storage log will start to appear in silo.log with "Storage" mentioned in the log message.

Make sure to enable storage logging only a moment before kicking off the collection process, and to disable it right after getting the logs since the log is really verbose and clogs up the logs otherwise.

Data Dumps

Sometimes, the data size is so large or the generated relationship data is so large that it exhausts all the memory space it's allocated. In that case, we need to look into the data to see why the result set is so large.

We would need to ask below table dump of customer data so we can load them onto the in-house system to reproduce the issue and look into data:

For L2 topology, below tables are needed:

master_dev.device_interfaces
master_dev.device_interfaces_map
master_dev.legend_device
master.definitions_dev_classes
master_dev.device_state
master_dev.device_ip_addr
master.definitions_dev_cats

If customer has concern about data table (especially for government customer), they can dump selected table columns using below steps to reduce the sharing of sensitive data.

This step will only need to have MAC address of interfaces. (We do need the mac address to perform L2 topology relationship building)

```
silo_mysql -u root -pem7admin master_dev -e 'SELECT if_id, did, roa_id, ifIndex, ifPhysAddress FROM master_dev.device_interfaces;' -X > if_table.xml
silo_mysql -u root -pem7admin master_dev -e 'SELECT id, roa_id, class_type FROM master_dev.legend_device;' -X > dev_table.xml
silo_mysql -u root -pem7admin master_dev -e 'SELECT did, state FROM master_dev.device_state;' -X > dstate_table.xml
silo_mysql -u root -pem7admin master_dev -e 'SELECT id, net_id, did, if_id, roa_id FROM master_dev.device_ip_addr;' -X > dip_table.xml
silo_mysql -u root -pem7admin master_dev -e 'SELECT * FROM master_dev.device_interfaces_map;' -X > dim_table.xml
silo_mysql -u root -pem7admin master_dev -e 'SELECT class_type, family_guid FROM master.definitions_dev_classes;' -X > dc_table.xml
```

We can then load these files using mysql terminal by:

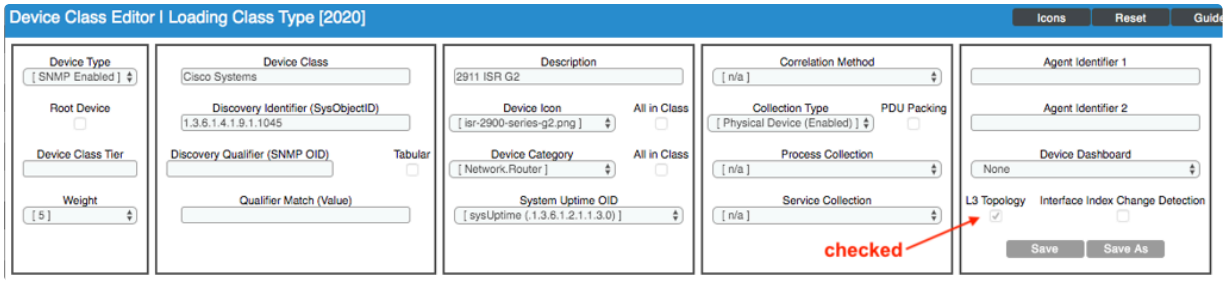
```
load xml local infile 'if_table.xml' into table master_dev.device_interfaces;
load xml local infile 'dev_table.xml' into table master_dev.legend_device;
load xml local infile 'dstate_table.xml' into table master_dev.device_state;
load xml local infile 'dip_table.xml' into table master_dev.device_ip_addr;
load xml local infile 'dim_table.xml' into table master_dev.device_interfaces_map;
load xml local infile 'dc_table.xml' into table master.definitions_dev_classes;
```

Theory

L3 Topology

To let L3 topology work, it needs to meet following requirements:

1. Run traceroute cmd from the collector against the target device. The target device and the aligned Collector (or All-in-One) appliance need to be discovered under the same collector group(CUG).
2. All IPs shown on the traceroute result need to be discovered on system under the same CUG. If an IP is a device's primary IP, it will use that device ID as one hop, otherwise, it will check the master_dev.device_ip_addr table for the first record it gets. (L3 topology record will not be created when any the device on the list is unknown)
3. The target device needs to have L3 collector flag enabled for its aligned device class



4. For Phone home collector (PHC) environment (EM-33560), if intending to utilize Layer 3 Topology for any devices monitored via a Phonehome collector, that collector must be discovered and inventoried via its localhost (127.0.0.x) IP address used at the Appliance Manager page. If the Phonehome collector has already been discovered/inventoried via a different IP address, the localhost IP address can be manually added to it via the Device Properties page (wrench icon). Either way, the IP address stored should match the IP address shown for that collector in the Appliances page.

Here is a sample debug log when L3 is working.

1. Run traceroute cmd on the Collector manually against the target device IP: 10.168.48.49

```
$traceroute 10.168.48.49


traceroute to 10.168.48.49 (10.168.48.49), 30 hops max, 60 byte packets
 1  10.2.9.5 (10.2.9.5)  0.564 ms  0.660 ms  0.789 ms
 2  10.128.3.9 (10.128.3.9)  0.577 ms  3.043 ms  3.275 ms
 3  t-edge.dc2.corp.sciencelogic.com (10.128.3.38)  2.708 ms  2.737 ms  2.777 ms
 4  10.64.3.12 (10.64.3.12)  4.057 ms t-edge.rst.corp.sciencelogic.com (10.64.3.4)  4.078 ms
4.104 ms
 5  core01.rst.corp.sciencelogic.com (10.64.3.33)  2.061 ms core02.rst.corp.sciencelogic.com
(10.64.3.37)  1.688 ms  1.727 ms
 6  core01.rst.eng.sciencelogic.com (10.64.131.17)  2.097 ms core02.rst.eng.sciencelogic.com
(10.64.131.19)  1.413 ms core01.rst.eng.sciencelogic.com (10.64.131.17)  2.061 ms
 7  10.150.0.3 (10.150.0.3)  1.272 ms  1.726 ms  1.893 ms
 8  10.168.48.93 (10.168.48.93)  1.291 ms  1.340 ms  1.767 ms
 9  10.168.48.49 (10.168.48.49)  1.986 ms * *
```

2. Run the L3 collection process manually on the Collector, and debug log shows the traceroute result, nothing fancy so far.

```
"2019-11-12 18:52:26,064","DEBUG","5711","34.PoolWorker-2","logger","195","Returning EXISTING
logger instance."
"2019-11-12 18:52:26,064","DEBUG","5711","34.PoolWorker-2","collect_l3topo","58","Starting l3
collect for ip 10.168.48.49"
"2019-11-12 18:52:26,685","DEBUG","5412","34","shared","164","got results for: 9"
"2019-11-12 18:52:26,685","DEBUG","5412","34","shared","164","got results for: 3"
"2019-11-12 18:52:27,585","DEBUG","5710","34.PoolWorker-1","collect_l3topo","65","Received
pickled object [{'hop': 1, 'rtt_time': 0.009361028671264648, 'address': '10.2.9.5'}, {'hop':
2, 'rtt_time': 0.0018939971923828125, 'address': '10.128.3.9'}, {'hop': 3, 'rtt_time':
0.0038220882415771484, 'address': '10.128.3.38'}, {'hop': 4, 'rtt_time':
0.004900932312011719, 'address': '10.64.3.4'}, {'hop': 5, 'rtt_time': 0.0018229484558105469,
'address': '10.64.3.37'}, {'hop': 6, 'rtt_time': 0.0014710426330566406, 'address':
'10.64.131.19'}, {'hop': 7, 'rtt_time': 0.0020699501037597656, 'address': '10.150.0.3'},
```

```
{'hop': 8, 'rtt_time': 0.001967191696166992, 'address': '10.168.48.93'}] when doing l3 collect for ip 10.168.48.93"
```

3. Then during L3 raw data storage, the storage log will tell why it may not store the result.

- When Collector is not discovered:

```
"2019-11-12 18:47:56,226","DEBUG","30955","Storage","storage_objects","2178","Store method called for l3_topology_storage"
"2019-11-12 18:47:56,242","DEBUG","30955","Storage","data_access","231","Collector id: 1 is not discovered, cannot store any L3 topology information."
```

- when some ip in the traceroute result is not discovered:

```
"2019-11-12 18:52:29,704","DEBUG","5412","Storage","storage_objects","2178","Store method called for l3_topology_storage"
"2019-11-12 18:52:29,727","DEBUG","5412","Storage","data_access","255","L3 Topo - Undiscovered device in path: 10.64.3.12. Cannot continue"

```

LLDP Topology

To let LLDP topology work, it needs to meet following requirements:

1. LLDP must be turned on in SL1 settings > Behavior page

2. The target device needs to be of certain device categories (network switches or network routers).

You could run the below query on the device's aligned Collector and make sure the result includes the device.

```
SELECT DISTINCT dev.did, dev.snmp_cred_id, dev.ip, dev.hostname
FROM collector_state.V_device as dev
JOIN master.definitions_dev_classes as dev_classes USING(class_type)
JOIN master.definitions_dev_cats as dev_cats ON (dev_cats.Fid = dev_classes.family)
WHERE (dev.active & 11) = 0 AND dev_cats.family_guid IN("6694D505DBA82BE89A7C4EE47DBA68B7", "0DC233A278BB81DFD37E944A96BDCE3D")
 AND dev.ip != '' AND dev.ip IS NOT NULL
```

Test devices

Layer 2: 10.140.235.72 - 10.140.235.91 (using SNMP Public V2 credential)

CDP: 10.140.235.92-10.140.235.99 (using SNMP Public V2 credential)

LLDP: 10.140.235.92-10.140.235.99 (using SNMP Public V2 credential)

Layer 3: depends on test system, but you can run traceroute against any discovered device and discover all IPs mentioned in the traceroute result.

(CBQOS: 10.140.235.94-10.140.235.96 using SNMP Public V2 credential)

Silo Log Messages

Introduction

Until this is better fleshed out, the reference example is here: [Silo Log Messages]

analysis script

This script, when run against silo.log files gathered with debug enabled for the process "Enterprise Database: Topology Crunch" (/opt/em7/backend/topology_crunch.py):

Error loading the extension!

will break the repetitive lines out into separate r_X.txt files, leaving only major checkpoint messages covering an entire run of the process in the last rN.txt file. If nothing else, check the last rN.txt file from the script for errors of unusual events.

Log Pattern Analysis

single:multinode_conn

```
"2020-02-11 06:25:29,017","DEBUG","24140","83","l2","524","Multi node connection:
{'br_if_id': 181149, 'br_did': 8764, 'vlan': 1, 'br_port_id': 5004, 'node_name': 'MXREP-CORE-
01.net.teletech.com', 'node_if_id': 182439, 'node_did': 8782, 'node_port_id': 293,
'node_mac': '00:21:56:e9:f0:00', 'net_id': 6198}"
"2020-02-11 06:25:29,033","DEBUG","24140","83","l2","524","Multi node connection:
{'br_if_id': 181149, 'br_did': 8764, 'vlan': 1, 'br_port_id': 5004, 'node_name': 'MXREP-CORE-
01.net.teletech.com', 'node_if_id': 182439, 'node_did': 8782, 'node_port_id': 293,
'node_mac': '00:21:56:e9:f0:00', 'net_id': 6199}"
"2020-02-11 06:25:29,033","DEBUG","24140","83","l2","524","Multi node connection:
{'br_if_id': 181149, 'br_did': 8764, 'vlan': 1, 'br_port_id': 5004, 'node_name': 'MXREP-CORE-
01.net.teletech.com', 'node_if_id': 182439, 'node_did': 8782, 'node_port_id': 293,
'node_mac': '00:21:56:e9:f0:00', 'net_id': 6200}"
```

stanza:node_connection

```
"2020-02-11 06:25:32,026","DEBUG","24140","83","l2","499","Single node connection:"
"2020-02-11 06:25:32,027","DEBUG","24140","83","shared","176","Attempting to add connection
for ((<Node id:8782 name:MXREP-CORE-01.net.teletech.com
```

```
family:0DC233A278BB81DFD37E944A96BDCE3D>,
"2020-02-11 06:25:32,027","DEBUG","24140","83","shared","183","Neither port is connected, so I will connect them."
"2020-02-11 06:25:32,027","DEBUG","24140","83","shared","153","ADDED CONNECTION: (<Node id:8782 name:MXREP-CORE-01.net.teletech.com family:0DC233A278BB81DFD37E944A96BDCE3D>, 247218, 1) to (<Node id:8778 name:MXREP-EDGE-01.net.teletech.com family:0>, 181980, 1) at weight 2"
"2020-02-11 06:25:32,027","DEBUG","24140","83","l2","499","Single node connection:"
"2020-02-11 06:25:32,027","DEBUG","24140","83","shared","176","Attempting to add connection for ((<Node id:8782 name:MXREP-CORE-01.net.teletech.com family:0DC233A278BB81DFD37E944A96BDCE3D>,
"2020-02-11 06:25:32,028","DEBUG","24140","83","shared","183","Neither port is connected, so I will connect them."
"2020-02-11 06:25:32,028","DEBUG","24140","83","shared","153","ADDED CONNECTION: (<Node id:8782 name:MXREP-CORE-01.net.teletech.com family:0DC233A278BB81DFD37E944A96BDCE3D>, 182298, 1) to (<Node id:13070 name:MXREP-EDGE-02.net.teletech.com family:0>, 180505, 1) at weight 2"
"2020-02-11 06:25:32,028","DEBUG","24140","83","l2","499","Single node connection:"
"2020-02-11 06:25:32,028","DEBUG","24140","83","shared","176","Attempting to add connection for ((<Node id:8782 name:MXREP-CORE-01.net.teletech.com family:0DC233A278BB81DFD37E944A96BDCE3D>,
"2020-02-11 06:25:32,028","DEBUG","24140","83","shared","183","Neither port is connected, so I will connect them."
"2020-02-11 06:25:32,028","DEBUG","24140","83","shared","153","ADDED CONNECTION: (<Node id:8782 name:MXREP-CORE-01.net.teletech.com family:0DC233A278BB81DFD37E944A96BDCE3D>, 182308, 1) to (<Node id:0 name:Unknown node family:00000000000000000000000000000000>, 197123, 1) at weight 2"
```

stanza:bridge_connection

```
"2020-02-11 08:17:24,290","DEBUG","18254","83","l2","299","Finding bridge half connections for USSPR-ACCESS-28.teletech.com - (8215)"
 "2020-02-11 08:17:24,293","DEBUG","18254","83","l2","302","-- Checking connections for [8216, 8217, 8218, 8219, ... 8183, 8184, 8185, 8186, 8187, 8188]"
"2020-02-11 08:17:24,509","DEBUG","18254","83","l2","299","Finding bridge half connections for USSPR-ACCESS-26.teletech.com - (8216)"
"2020-02-11 08:17:24,516","DEBUG","18254","83","l2","302","-- Checking connections for [8215, 8217, 8218, 8219, ... 8185, 8186, 8187, 8188]"
```

single:did_sees_did

```
"2020-02-11 08:17:33,831","DEBUG","18254","83","l2","313","-- DID 8251 sees DID 8311 on FDBPort 5001 VLAN 1"
"2020-02-11 08:17:33,832","DEBUG","18254","83","l2","313","-- DID 8253 sees DID 8311 on FDBPort 5001 VLAN 1"
"2020-02-11 08:17:33,832","DEBUG","18254","83","l2","313","-- DID 8262 sees DID 8311 on FDBPort 5001 VLAN 1"
```

stanza:pass_fail

```
"2020-02-11 08:23:09,230","DEBUG","18254","83","l2","334","Connection (7950, 13067, 1) passes pre-test #1"
"2020-02-11 08:23:09,231","DEBUG","18254","83","l2","350","Connection (7950, 13067, 1) passes test #1"
"2020-02-11 08:23:09,231","DEBUG","18254","83","l2","336","Connection (7826, 7804, 1) failed pre-test #1, removing"
```

```
"2020-02-11 08:23:09,232","DEBUG","18254","83","l2","336","Connection (8953, 10931, 1) failed pre-test #1, removing"
"2020-02-11 08:23:09,232","DEBUG","18254","83","l2","334","Connection (8010, 8072, 1) passes pre-test #1"
"2020-02-11 08:23:09,233","DEBUG","18254","83","l2","352","Connection (8010, 8072, 1) failed test #1, removing"
"2020-02-11 08:23:09,234","DEBUG","18254","83","l2","336","Connection (8815, 8845, 1) failed pre-test #1, removing"
"2020-02-11 08:23:09,234","DEBUG","18254","83","l2","336","Connection (8589, 8421, 1) failed pre-test #1, removing"
"2020-02-11 08:23:09,234","DEBUG","18254","83","l2","336","Connection (8832, 8896, 1) failed pre-test #1, removing"
"2020-02-11 08:23:09,235","DEBUG","18254","83","l2","334","Connection (8459, 8512, 1) passes pre-test #1"
"2020-02-11 08:23:09,236","DEBUG","18254","83","l2","350","Connection (8459, 8512, 1) passes test #1"
```

stanza:port_connections

```
"2020-02-11 08:23:32,059","DEBUG","18254","83","shared","187","(<Node id:59851 name:USDEN-DIST-08.net.teletech.com family:0DC233A278BB81DFD37E944A96BDCE3D>, 402541, 1): 23 (<Node id:11119 name:USDEN-DIST-07.net.teletech.com family:0DC233A278BB81DFD37E944A96BDCE3D>, 297986, 1): 23"
"2020-02-11 08:23:32,060","DEBUG","18254","83","shared","191","Ports are connected to each other at 2 weight, new weight is 2"
"2020-02-11 08:23:32,063","DEBUG","18254","83","shared","187","(<Node id:8798 name:BGSOF-ACCESS-03.net.teletech.com family:0DC233A278BB81DFD37E944A96BDCE3D>, 183437, 1): 14 (<Node id:8935 name:BGSOF-CORE-02.teletech.net family:0DC233A278BB81DFD37E944A96BDCE3D>, 192527, 1): 14"
"2020-02-11 08:23:32,063","DEBUG","18254","83","shared","191","Ports are connected to each other at 2 weight, new weight is 2"
"2020-02-11 08:23:32,069","DEBUG","18254","83","shared","187","(<Node id:8885 name:BGSOF-CORE-01.teletech.net family:0DC233A278BB81DFD37E944A96BDCE3D>, 321452, 1): 37 (<Node id:8796 name:BGSOF-ACCESS-05.net.teletech.com family:0DC233A278BB81DFD37E944A96BDCE3D>, 183261, 1): 37"
"2020-02-11 08:23:32,069","DEBUG","18254","83","shared","191","Ports are connected to each other at 2 weight, new weight is 2"
```

remainder

The messages below are what's left after all of the repetitive single and multiple line stanzas have been removed. You would expect what's left to have major milestones of process execution and rare events.

```
"2020-02-11 08:01:02,434","DEBUG","18254","83","proc_pda","256","Got proc virtual memory limit from silo.conf:None"
"2020-02-11 08:01:02,434","DEBUG","18254","83","proc_pda","268","No or Incorrect virtual memory limit from silo.conf:proc_id=83 : None"
"2020-02-11 08:01:02,453","DEBUG","18254","83","proc_pda","190","Logging started for Enterprise Database: Topology Crunch"
"2020-02-11 08:01:02,454","DEBUG","18254","83","proc_pda","500","Started tracking process: 254 83 a55b4bc24ca411ea81de005056816592 "
"2020-02-11 08:01:02,454","DEBUG","18254","83","proc_pda","374","Trying to get system settings"
"2020-02-11 08:01:02,456","DEBUG","18254","83","crunch_topology","55","Topology type l2 (crunch) is enabled"
```

```
"2020-02-11 08:01:02,456","DEBUG","18254","83","crunch_topology","73","Crunching Topology for type l2"
"2020-02-11 08:17:23,510","DEBUG","18254","83","shared","118","Loading override connections for 2"
"2020-02-11 08:17:24,261","DEBUG","18254","83","l2","279","Loaded 6151 nodes"
"2020-02-11 08:17:24,269","DEBUG","18254","83","l2","280","Loaded 1487 bridges"
"2020-02-11 08:17:24,269","DEBUG","18254","83","shared","118","Loading override connections for 2"
"2020-02-11 08:23:09,135","DEBUG","18254","83","l2","316","After Step 1. Half connections:"
"2020-02-11 08:23:09,229","DEBUG","18254","83","l2","317","
"2020-02-11 08:23:31,609","DEBUG","18254","83","l2","356","After Test #1. Half connections:"
"2020-02-11 08:23:31,620","DEBUG","18254","83","l2","357","
"2020-02-11 08:23:31,861","DEBUG","18254","83","l2","401","After Test #2. Half connections:"
"2020-02-11 08:23:31,868","DEBUG","18254","83","l2","402","
"2020-02-11 08:23:31,997","DEBUG","18254","83","l2","445","After Test 3. Half connections:"
"2020-02-11 08:23:32,003","DEBUG","18254","83","l2","446","
"2020-02-11 08:23:32,268","DEBUG","18254","83","shared","200","Both Ports are connected but not to each other. weight1: 2 weight2: 2 new weight: 2"
"2020-02-11 08:23:32,270","DEBUG","18254","83","shared","200","Both Ports are connected but not to each other. weight1: 2 weight2: 2 new weight: 2"
"2020-02-11 08:23:32,324","DEBUG","18254","83","shared","200","Both Ports are connected but not to each other. weight1: 2 weight2: 2 new weight: 2"
"2020-02-11 08:23:32,400","DEBUG","18254","83","shared","200","Both Ports are connected but not to each other. weight1: 2 weight2: 2 new weight: 2"
"2020-02-11 08:23:32,423","DEBUG","18254","83","l2","464","Full bridge connections:"
"2020-02-11 08:23:32,440","DEBUG","18254","83","l2","465","
"2020-02-11 08:24:22,839","DEBUG","18254","83","proc_pda","528","tracking update: 254 83 a55b4bc24ca411ea81de005056816592 "
"2020-02-11 08:24:22,840","DEBUG","18254","83","proc_pda","545","Stopped tracking process: 254 83 a55b4bc24ca411ea81de005056816592 "
"2020-02-11 08:24:22,840","ERROR","18254","83","proc_pda","306","MemoryError Exception, exiting"
"2020-02-11 08:24:22,841","DEBUG","18254","83","proc_pda","307","Logging stopped for Enterprise Database: Topology Crunch"
"2020-02-11 10:01:02,382","DEBUG","11719","83","proc_pda","256","Got proc virtual memory limit from silo.conf:None"
"2020-02-11 10:01:02,382","DEBUG","11719","83","proc_pda","268","No or Incorrect virtual memory limit from silo.conf:proc_id=83 : None"
"2020-02-11 10:01:02,411","DEBUG","11719","83","proc_pda","190","Logging started for Enterprise Database: Topology Crunch"
"2020-02-11 10:01:02,412","DEBUG","11719","83","proc_pda","500","Started tracking process: 254 83 68d7a04a4cb511eab6c2005056816592 "
"2020-02-11 10:01:02,413","DEBUG","11719","83","proc_pda","374","Trying to get system settings"
"2020-02-11 10:01:02,415","DEBUG","11719","83","crunch_topology","55","Topology type l2 (crunch) is enabled"
"2020-02-11 10:01:02,415","DEBUG","11719","83","crunch_topology","73","Crunching Topology for type l2"
"2020-02-11 10:22:19,339","DEBUG","11719","83","shared","118","Loading override connections for 2"
"2020-02-11 10:22:20,159","DEBUG","11719","83","l2","279","Loaded 6151 nodes"
"2020-02-11 10:22:20,167","DEBUG","11719","83","l2","280","Loaded 1487 bridges"
"2020-02-11 10:22:20,167","DEBUG","11719","83","shared","118","Loading override connections for 2"
"2020-02-11 10:28:07,489","DEBUG","11719","83","l2","316","After Step 1. Half connections:"
"2020-02-11 10:28:07,608","DEBUG","11719","83","l2","317","
"2020-02-11 10:28:30,277","DEBUG","11719","83","l2","356","After Test #1. Half connections:"
"2020-02-11 10:28:30,290","DEBUG","11719","83","l2","357","
```

```
"2020-02-11 10:28:30,510","DEBUG","11719","83","l2","401","After Test #2. Half connections:"
"2020-02-11 10:28:30,517","DEBUG","11719","83","l2","402","
"2020-02-11 10:28:30,656","DEBUG","11719","83","l2","445","After Test 3. Half connections:"
"2020-02-11 10:28:30,663","DEBUG","11719","83","l2","446","
"2020-02-11 10:28:30,887","DEBUG","11719","83","shared","200","Both Ports are connected but not to each other. weight1: 2 weight2: 2 new weight: 2"
"2020-02-11 10:28:30,888","DEBUG","11719","83","shared","200","Both Ports are connected but not to each other. weight1: 2 weight2: 2 new weight: 2"
"2020-02-11 10:28:30,938","DEBUG","11719","83","shared","200","Both Ports are connected but not to each other. weight1: 2 weight2: 2 new weight: 2"
"2020-02-11 10:28:31,022","DEBUG","11719","83","shared","200","Both Ports are connected but not to each other. weight1: 2 weight2: 2 new weight: 2"
"2020-02-11 10:28:31,055","DEBUG","11719","83","l2","464","Full bridge connections:"
"2020-02-11 10:28:31,072","DEBUG","11719","83","l2","465","
```

Above we have two process executions (18254, followed by 11719). You would normally expect an execution to start and end with:

```
"2020-02-11 08:01:02,434","DEBUG","18254","83","proc_pda","256","Got proc virtual memory limit from silo.conf:None"
"2020-02-11 08:01:02,434","DEBUG","18254","83","proc_pda","268","No or Incorrect virtual memory limit from silo.conf:proc_id=83 : None"
"2020-02-11 08:01:02,453","DEBUG","18254","83","proc_pda","190","Logging started for Enterprise Database: Topology Crunch"
...
"2020-02-11 08:24:22,841","DEBUG","18254","83","proc_pda","307","Logging stopped for Enterprise Database: Topology Crunch"
```

Lines here that would seem like outliers, and should be investigated:

```
"2020-02-11 08:23:32,268","DEBUG","18254","83","shared","200","Both Ports are connected but not to each other. weight1: 2 weight2: 2 new weight: 2"
"2020-02-11 08:23:32,270","DEBUG","18254","83","shared","200","Both Ports are connected but not to each other. weight1: 2 weight2: 2 new weight: 2"
"2020-02-11 08:23:32,324","DEBUG","18254","83","shared","200","Both Ports are connected but not to each other. weight1: 2 weight2: 2 new weight: 2"
"2020-02-11 08:23:32,400","DEBUG","18254","83","shared","200","Both Ports are connected but not to each other. weight1: 2 weight2: 2 new weight: 2"
...
"2020-02-11 08:24:22,840","ERROR","18254","83","proc_pda","306","MemoryError Exception, exiting"
...
```

Consumer of Topology Relationship

[From Topology Relationships to Event suppression]

Post Mortem Articles

2024-08-27 EM-66573: Single Digits - customization of topology - drop file install for testing

ts_guide_to_jira_link

Link: 💔EM-66573: Single Digits - customization of topology - drop file install for testing TO DO

From the description:

```
Observed Behavior: Customer would like some customization to topology crunch. Product/Eng team agreed to provide drop files to demonstrate capabilities.

Expected Behavior: N/A

Summary of troubleshooting steps taken so far and the current status:
Randy Murdock has details about the drop file request.
Thank you.
-Leo
```

2024-08-13 EM-66384: EM7 minor event: Storage Object Failure due to ERROR. Type: lldp_topology_storage

ts_guide_to_jira_link

Link: EM-66384: EM7 minor event: Storage Object Failure due to ERROR. Type: lldp_topology_storage CLOSED

From the description:

```
Observed Behavior: Hundreds of lldp_topology_storage sigterms with unicode errors are appearing in the system logs

Expected Behavior: Fix the errors

Summary of troubleshooting steps taken so far and the current status:
Customer reached out stating that they are getting below events:

"EM7 minor event: Storage Object Failure due to ERROR. Type: lldp_topology_storage DID: 1183 Policy Info: lldp_topology_storage Collector ID: 9 Row ID: 95948491 Additional Error Information:
Exception UnicodeEncodeError Traceback: Traceback (most recent call last): File"

Here is the screenshot of the error:

The above screenshot shows the same error as mentioned in below escalation:
https://sciencelogic.atlassian.net/browse/EM-64633

DB Recent sigterms:

Logs from DB:

Now the issue here is the logs were captured when the event got cleared.
```

```

However, customer mentioned that this has been a transient issue for a while and the error might show up again.

In escalation:
https://sciencelogic.atlassian.net/browse/EM-64633

I could see a drop file was provided which fixed the issue for customer.

Questions:
Can we get drop file for this stack?
```

The screenshot in the description (not quoted above) shows this is another example of a exception where:

```
'charmap' codec can't encode character UHE
```

2024-08-01 EM-66229: cdp_collect.py throwing exceptions over bad neighbor IP addresses

ts_guide_to_jira_link

Link: 💔 EM-66229: cdp_collect.py throwing exceptions over bad neighbor IP addresses CLOSED

From the description:

```
Observed Behavior:
Dropfile applied to all collectors to get rid of cdp_collect.py UHEs in case 00449615/https://sciencelogic.atlassian.net/browse/EM-65721#icft=EM-65721 - Unhandled exception: CDP Collection Process
Now cdp_collect.py is throwing exceptions claiming it's getting a blank IP, however, an SNMP walk of the MIB does not show any blank values:

2024-07-23T18:11:08.405821+00:00 localhost silo.33: MainProcess[4190051]: ERROR collect_cdp.collect_cdp_neighbors 103 Exception caught while converting IP address of neighbor of DID 17537 at IP 10.128.255.75: #012
Traceback (most recent call    last):#012
File "collect_cdp.py", line 100, in collect_cdp_neighbors#012
File "/opt/em7/lib/python3.6/network.py", line 293, in from_hexip#012
File "/opt/em7/lib/python3.6/network.py", line 120, in _init_#012sl_network.exceptions.IPAddressError: Address passed was not valid.

Running the cdp_collect process in debug mode yields the following:
2024-07-23T18:11:08.405544+00:00 localhost silo.33: MainProcess[4190051]: DEBUG network.from_hexip 272
2024-07-23T18:11:08.405579+00:00 localhost silo.33: MainProcess[4190051]: DEBUG network.from_hexip 279 Making value printable
2024-07-23T18:11:08.405619+00:00 localhost silo.33: MainProcess[4190051]: DEBUG network.from_hexip 283 Converted IP to hex list: ([],)
2024-07-23T18:11:08.405658+00:00 localhost silo.33: MainProcess[4190051]: DEBUG network.from_hexip 290 Converted IP to string:

Expected Behavior:
CDP collects properly.

Summary of troubleshooting steps taken so far and the current status:
```

```
Dropfile applied to all collectors to get rid of cdp_collect.py UHEs in case 00449615/https://sciencelogic.atlassian.net/browse/EM-65721#icft=EM-65721 - Unhandled exception: CDP Collection Process
Now cdp_collect.py is throwing exceptions claiming it's getting a blank IP, however, an SNMP walk of the MIB does not show any blank values
```

2024-07-30 EM-66192: UnicodeEncodeError: 'charmap' codec can't encode character UHE

ts_guide_to_jira_link

Link: EM-66192: UnicodeEncodeError: 'charmap' codec can't encode character UHE CLOSED

From the description:

```
Observed Behavior: The em7_hfpulld process is failing with UHE UnicodeEncodeError: 'charmap' codec can't encode character.

Expected Behavior: Error should not be generating.

Summary of troubleshooting steps taken so far and the current status:
Customer reports they are unable to discover interfaces due to the following UHE's.
silo.em7_mfpulld.78: Storage Object Failure due to ERROR. Type: lldp_topology_storage DID: 5632 Policy Info: lldp_topology_storage Collector ID: 10 Row ID: 235679177 Additional Error Information: Exception UnicodeEncodeError Traceback:
Traceback (most recent call last):
File "/opt/em7/lib/python3.6/storage_process.py", line 231, in store_object
File "/opt/em7/lib/python3.6/storage_objects.py", line 98, in store
File "/opt/em7/lib/python3.6/storage_objects.py", line 106, in store_neighbors
File "/opt/em7/lib/python3.6/data_access.py", line 304, in store_lldp_neighbors
File "/opt/em7/lib/python3.6/data_access.py", line 107, in did_from_chassis
File "/opt/em7/lib/python3.6/cursor.py", line 656, in autofetch_value
File "/opt/em7/lib/python3.6/cursor.py", line 120, in wrapper
File "/opt/em7/lib/python3.6/cursor.py", line 99, in callFunc
File "/opt/em7/lib/python3.6/cursor.py", line 464, in execute
File "/opt/em7/lib/python3/MySQLdb/cursors.py", line 199, in execute args = tuple(map(db.literal, args))
File "/opt/em7/lib/python3/MySQLdb/connections.py", line 275, in literal s = self.string_literal(o.encode(self.encoding))
File "/usr/lib64/python3.6/encodings/cp1252.py", line 12, in encode return codecs.charmap_encode(input,errors,encoding_table) UnicodeEncodeError: 'charmap' codec can't encode character '\x90' in position 4: character maps to <undefined>

silo.em7_mfpulld.78: Storage Object Failure due to ERROR. Type: lldp_topology_storage DID: 4819 Policy Info: lldp_topology_storage Collector ID: 3 Row ID: 232643452 Additional Error Information: Exception UnicodeEncodeError
Traceback: Traceback (most recent call last):
File "/opt/em7/lib/python3.6/storage_process.py", line 231, in store_object
File "/opt/em7/lib/python3.6/storage_objects.py", line 98, in store
File "/opt/em7/lib/python3.6/storage_objects.py", line 106, in store_neighbors
File "/opt/em7/lib/python3.6/data_access.py", line 348, in store_lldp_neighbors
File "/opt/em7/lib/python3.6/cursor.py", line 99, in callFunc
File "/opt/em7/lib/python3.6/cursor.py", line 640, in executemany
File "/opt/em7/lib/python3/MySQLdb/cursors.py", line 236, in executemany self._get_db().encoding,
File "/opt/em7/lib/python3/MySQLdb/cursors.py", line 259, in _do_execute_many v = values % escape(arg, conn)
```

```
File "/opt/em7/lib/python3/MySQLdb/cursors.py", line 110, in _escape_args ret = tuple(literal(ensure_bytes(arg)) for arg in args)
File "/opt/em7/lib/python3/MySQLdb/cursors.py", line 110, in <genexpr> ret = tuple(literal(ensure_bytes(arg)) for arg in args)
File "/opt/em7/lib/python3/MySQLdb/cursors.py", line 102, in ensure_bytes return x.encode(encoding)
File "/usr/lib64/python3.6/encodings/cp1252.py", line 12, in encode return codecs.charmap_encode(input,errors,encoding_table)
 UnicodeEncodeError: 'charmap' codec can't encode character '\x94' in position 0: character maps to <undefined>

The issue appears to be widespread and affects all collectors. This was requested to be escalated to Proc and Roll by Ben Leyland.
```

2024-07-08 EM-65721: Unhandled exception: CDP Collection Process

ts_guide_to_jira_link

Link: EM-65721: Unhandled exception: CDP Collection Process CLOSED

From the description:

```
Observed Behavior:
every 2 hours CDP Collections error with UHE

Expected Behavior:
Should collect without error

Summary of troubleshooting steps taken so far and the current status:
(Support Engineers need to provide a detailed summary of what steps have been taken so far and where they are currently at with the issue.)
I began researching the issue and there does not appear to be a knowledge article for this yet but this slack thread:
https://sciencelogic.slack.com/archives/C04L3KNQLGJ/p1710528994173919

Led me to this Jira
https://sciencelogic.atlassian.net/browse/EM-62605
Which I believe is the same issue. If this is true and it is the same issue, the customer wouldlike explicit instructions for applying the drop file as well as any guidance that can be provided as to
the most efficient way to apply the dorp file across ~40 collectors if such a method exists.
```

EM-62605 is linked in a section below.

The slack discussion is here: https://app.slack.com/client/E03R7NUFC4X/C04L3KNQLGJ

The UHE quoted in the slack discussion is here:

```
2024-03-08T20:01:07.497842+00:00 localhost silo.cdp_collect.33[3891538]: DEBUG shared._run_collect_sync 162 collect task for: 4113
2024-03-08T20:01:08.805926+00:00 localhost silo.cdp_collect.33[3891538]: CRITICAL proc_pda.__init__ 260 Unhandled exception on appliance: 3, process Data Collection: CDP Collection:#012Traceback (most recent call last):#012
File "/opt/em7/lib/python3.6/proc_pda.py", line 215, in __init__#012
File "/opt/em7/lib/python3.6/collect_cdp.py", line 163, in main#012
File "/opt/em7/lib/python3.6/shared.py", line 154, in run_collection#012
File "/opt/em7/lib/python3.6/shared.py", line 167, in _run_collect_sync#012
```

```
File "/opt/em7/lib/python3.6/collect_cdp.py", line 151, in collect#012
File "/opt/em7/lib/python3.6/collect_cdp.py", line 102, in collect_cdp_neighbors#012
File "/opt/em7/lib/python3.6/network.py", line 293, in from_hexip#012
File "/opt/em7/lib/python3.6/network.py", line 120, in
__init__#012sl_network.exceptions.IPAddressError: Address passed was not valid

2024-03-08T20:01:08.812982+00:00 localhost silo.cdp_collect.33[3891538]: DEBUG
proc_pda.update_tracker 796 tracking update: 3 33 75f65814dd8611ee808a005056871dbd
2024-03-08T20:01:08.815919+00:00 localhost silo.cdp_collect.33[3891538]: DEBUG
proc_pda._stop_tracker 818 Stopped tracking process: 3 33 75f65814dd8611ee808a005056871dbd
2024-03-08T20:01:08.816103+00:00 localhost silo.cdp_collect.33[3891538]: ERROR
proc_pda.close 462 Unhandled exception: Traceback (most recent call last):#012
File "/opt/em7/lib/python3.6/proc_pda.py", line 215, in __init__#012
File "/opt/em7/lib/python3.6/collect_cdp.py", line 163, in main#012
File "/opt/em7/lib/python3.6/shared.py", line 154, in run_collection#012
File "/opt/em7/lib/python3.6/shared.py", line 167, in _run_collect_sync#012
File "/opt/em7/lib/python3.6/collect_cdp.py", line 151, in collect#012
File "/opt/em7/lib/python3.6/collect_cdp.py", line 102, in collect_cdp_neighbors#012
File "/opt/em7/lib/python3.6/network.py", line 293, in from_hexip#012
File "/opt/em7/lib/python3.6/network.py", line 120, in
__init__#012sl_network.exceptions.IPAddressError: Address passed was not valid

2024-03-08T20:01:08.819619+00:00 localhost silo.cdp_collect.33[3891538]: DEBUG
proc_pda.close 463 Logging stopped for Data Collection: CDP Collection
```

2024-06-14 EM-65467: CDP Collection Error

ts_guide_to_jira_link

Link: EM-65467: Unhandled IPAddressError exception during CDP topology collection CLOSED

From the description:

```
Observed Behavior: It looks similar to https://sciencelogic.atlassian.net/browse/EM-59710
but for a different process:

EM7 critical event: Unhandled exception on appliance: 173, process Data Collection: CDP
Collection:
Traceback (most recent call last):
File "/opt/em7/lib/python3.6/proc_pda.py", line 215, in __init__
File "/opt/em7/lib/python3.6/collect_cdp.py", line 163, in main
File "/opt/em7/lib/python3.6/shared.py", line 154, in run_collection
File "/opt/em7/lib/python3.6/shared.py", line 167, in _run_collect_sync
File "/opt/em7/lib/python3.6/collect_cdp.py", line 151, in collect
File "/opt/em7/lib/python3.6/collect_cdp.py", line 102, in collect_cdp_neighbors
File "/opt/em7/lib/python3.6/network.py", line 293, in from_hexip
File "/opt/em7/lib/python3.6/network.py", line 120, in __init__
sl_network.exceptions.IPAddressError: Address passed was not valid

Expected Behavior:
No UHE

Summary of troubleshooting steps taken so far and the current status:
The customer says this began occurring after the drop file from
https://sciencelogic.atlassian.net/browse/EM-64382  was applied
Confirmed that it is not the issue documented in KB 9917 - the below query on the CBD
provided no results:
```

```
SELECT * FROM master_dev.device_ip_addr WHERE ip=' %' OR ip LIKE '% ' or netmask LIKE ' %' OR netmask LIKE '% ' OR (NOT IS_IPV4(ip)) AND (NOT IS_IPV6(ip));
I didn't consider this until now, the KB doesn't say where to run this query. Maybe I should have had the customer run this on the CU getting the UHE?
```

2024-06-12 EM-65416: Topology Collection not completing.

ts_guide_to_jira_link

Link: 💔EM-65416: Topology Collection hangs/does not complete, resulting in sigterms CLOSED

From the description:

```
Observed Behavior:
    Topology Collection appears to stall when attempting to collect information from a host that doesn't respond to "1.3.6.1.2.1.4.20.1.1" (ipAddrEntry) and results in the Topology Collection process sigterm'ing

Expected Behavior:
    No single device should cause a the topology collection to get stuck.

Summary of troubleshooting steps taken so far and the current status:
    Collected the topology collection debug logs from both collectors and noted that the collection always got stuck at a particular device, DID 4038 on MID 20 and DID 4039 on MID 24 in subsequent collections.
    The collector would always log "150 appending task to queue: 4038" but unlike working devices would never log "204 got results for: 4038", finally the collector would log "288 Data Collection: Topology task list at termination: [4038]".
    Attemted to increase the collection runtime (2 -> 4 hours) but this did not make a difference.
    Attempted to perform a snmpwalk of the ipAddrEntry ("1.3.6.1.2.1.4.20.1.1") on the "non working" device and noted that it responded that the OID did not exit or was empty.
    I'm assuming that the device id refered to below is DID 4038 and it's causing the collection to wait for it to respond.
    Attached are both the logs and full snmpwalks for both DID 4038 and 4039.

--snip--
2024-04-07T20:12:03.626982+00:00 localhost silo.topology_collect.25[9534]: DEBUG proc_pda.mem_limit_config 277 Got proc virtual memory limit from silo.conf:None
2024-04-07T20:12:03.627231+00:00 localhost silo.topology_collect.25[9534]: DEBUG proc_pda.mem_limit_config 289 No or Incorrect virtual memory limit from silo.conf:proc_id=25 : None
2024-04-07T20:12:03.636466+00:00 localhost silo.topology_collect.25[9534]: DEBUG proc_pda.init 198 Logging started for Data Collection: Topology
2024-04-07T20:12:03.636673+00:00 localhost silo.topology_collect.25[9534]: DEBUG proc_pda.init 223 importing pda tracker
2024-04-07T20:12:03.638128+00:00 localhost silo.topology_collect.25[9534]: DEBUG proc_pda._start_tracker 532 Started tracking process: 20 25 1983f65cf51b11eeaea80050560d04bd
2024-04-07T20:12:03.638274+00:00 localhost silo.topology_collect.25[9534]: DEBUG proc_pda.get_system_settings 406 Trying to get system settings
2024-04-07T20:12:03.639666+00:00 localhost silo.topology_collect.25[9534]: DEBUG shared.set_time 224 Official collection time: 2024-04-07 20:12:00 timestamp:1712520720
====truncated====
2024-04-07T20:44:20.887125+00:00 localhost silo.topology_collect.25[9534]: DEBUG shared._run_collect_async 204 got results for: 3445
2024-04-07T21:15:14.844163+00:00 localhost silo.topology_collect.25[9534]: DEBUG shared._run_collect_async 204 got results for: 18522
```

```
2024-04-07T21:15:19.849363+00:00 localhost silo.topology_collect.25[9534]: DEBUG shared._run_collect_async 204 got results for: 18548
2024-04-07T21:18:45.068505+00:00 localhost silo.topology_collect.25[9534]: DEBUG shared._run_collect_async 204 got results for: 18546
2024-04-07T23:34:55.741796+00:00 localhost silo.topology_collect.25[9534]: DEBUG shared._run_collect_async 204 got results for: 4251
2024-04-08T04:14:00.267683+00:00 localhost silo.topology_collect.25[9534]: DEBUG shared.shutdown 276 Data Collection: Topology.MainProcess has received a termination signal and is shutting down.
2024-04-08T04:14:00.267782+00:00 localhost silo.topology_collect.25[9534]: DEBUG shared.shutdown 281 Stopping poolworker processes
2024-04-08T04:15:00.738509+00:00 localhost silo.topology_collect.25[9534]: DEBUG shared.shutdown 276 Data Collection: Topology.MainProcess has received a termination signal and is shutting down.
2024-04-08T04:15:00.738587+00:00 localhost silo.topology_collect.25[9534]: DEBUG shared.shutdown 281 Stopping poolworker processes
2024-04-08T04:15:00.738696+00:00 localhost silo.topology_collect.25[9534]: INFO shared.shutdown 288 Data Collection: Topology task list at termination: [4038]
--snip--
```

2024-05-24 EM-65153: Storage Object Failure due to ERROR. Type: l3_topology_storage

ts_guide_to_jira_link

Link: EM-65153: Storage Object Failure due to ERROR. Type: l3_topology_storage CLOSED

From the description:

```
Observed Behavior:  We are seeing "Storage Object Failure due to ERROR. Type: l3_topology_storage on did;0"
Expected Behavior: This should run without an failure.
Summary of troubleshooting steps taken so far and the current status:
Collected the information from the requester.

EM7 minor event: Storage Object Failure due to ERROR. Type: l3_topology_storage DID: 0
Policy Info: l3_topology_storage Collector ID: 54 Row ID: 20799428 Additional Error Information: Exception ImportError Traceback: Traceback (most recent call last): File "/opt/em7/bac
```

2024-04-25 EM-64633: LLDP Topology Storage Errors

ts_guide_to_jira_link

Link: EM-64633: LLDP Topology Storage Errors - 'charmap' codec can't encode character CLOSED

From the description:

```
Observed Behavior: Hundreds of lldp_topology_storage sigterms with unicode errors are appearing in the system logs
Expected Behavior: lldp_topology_storage should not crash.

Summary of troubleshooting steps taken so far and the current status:
Collected logs from the DB and Collector
Identified sigterm from lldp_topology_storage causing unicode errors. Other similar unicode errors have been found in the logs.
```

```

Error message seen: from lldp topology:
1   silo.em7_mfpulld.78: Storage Object Failure due to ERROR. Type: lldp_topology_storage DID: 12612 Policy Info: lldp_topology_storage Collector ID: 56 Row ID: 15829832\nAdditional Error Information: Exception UnicodeEncodeError Traceback:\n
Traceback (most recent call last):\n
File "/opt/em7/lib/python3.6/storage_process.py", line 231, in store_object\n
File "/opt/em7/lib/python3.6/storage_objects.py", line 98, in store\n
File "/opt/em7/lib/python3.6/storage_objects.py", line 106, in store_neighbors\n
File "/opt/em7/lib/python3.6/data_access.py", line 304, in store_lldp_neighbors\n
File "/opt/em7/lib/python3.6/data_access.py", line 107, in did_from_chassis\n
File "/opt/em7/lib/python3.6/cursor.py", line 656, in autofetch_value\n
File "/opt/em7/lib/python3.6/cursor.py", line 120, in wrapper\n
File "/opt/em7/lib/python3.6/cursor.py", line 99, in callFunc\n
File "/opt/em7/lib/python3.6/cursor.py", line 464, in execute\n
File "/opt/em7/lib/python3/MySQLdb/cursors.py", line 199, in execute\n    args = tuple(map(db.literal, args))\n
File "/opt/em7/lib/python3/MySQLdb/connections.py", line 275, in literal\n    s = self.string_literal(o.encode(self.encoding))\n
File "/usr/lib64/python3.6/encodings/cp1252.py", line 12, in encode\n    return codecs.charmap_encode(input,errors,encoding_table)\nUnicodeEncodeError: 'charmap' codec can't encode character '\\x93' in position 4: character maps to <undefined>\n  2024-03-23 06:33:13 2100476

Other unicode errors were also found that were not directly related:
1   silo.77:StorageProc20: Unhandled exception in storage: \nTraceback (most recent call last):\n
File "/opt/em7/lib/python3.6/discovery_storage.py", line 2988, in store\n
File "/opt/em7/lib/python3.6/discovery_storage.py", line 2142, in update_device\n
File "/opt/em7/lib/python3.6/discovery_storage.py", line 2466, in update_interfaces\n
File "/opt/em7/lib/python3.6/interface.py", line 1245, in update\n
File "/opt/em7/lib/python3.6/interface.py", line 867, in _insert_as_new\n
File "/opt/em7/lib/python3.6/cursor.py", line 120, in wrapper\n
File "/opt/em7/lib/python3.6/cursor.py", line 99, in callFunc\n
File "/opt/em7/lib/python3.6/cursor.py", line 464, in execute\n
File "/opt/em7/lib/python3/MySQLdb/cursors.py", line 199, in execute\n    args = tuple(map(db.literal, args))\n
File "/opt/em7/lib/python3/MySQLdb/connections.py", line 275, in literal\n    s = self.string_literal(o.encode(self.encoding))\n
File "/usr/lib64/python3.6/encodings/cp1252.py", line 12, in encode\n    return codecs.charmap_encode(input,errors,encoding_table)\nUnicodeEncodeError: 'charmap' codec can't encode characters in position 19-20: character maps to <undefined>    2024-03-25 02:06:49    40

```

2024-04-04 EM-64160: DXC Technology Staging SL1 is generating SL1 internal errors

ts_guide_to_jira_link

Link: 💔EM-64160: DXC Technology Staging SL1 is generating SL1 internal errors around a device's invalid IP address CLOSED

From the description:

```
Observed Behavior: Customer is getting Unhandled Exceptions during CDP topology discovery.
Expected Behavior: There should be no Unhandled Exceptions. The customer requires CDP topology for their Cisco devices.
```

```
Summary of troubleshooting steps taken so far and the current status:
Please create a drop file for EM-62605.
Original description:

DXC Technology Staging SL1 is generating SL1 internal errors.

TOP 10 UNHANDLED EXCEPTION COUNTS
LOG MESSAGE COUNT
silo.cdp_collect.33: Unhandled exception: Traceback (most recent call last): File "/opt/em7/lib/python3.6/proc_pda.py", line 215, in _init_ File "/opt/em7/lib/python3.6/collect_cdp.py", line 163, in main File "/opt/em7/lib/python3.6/shared.py", line    336
silo.cdp_collect.33: Unhandled exception on appliance: 3, process Data Collection: CDP Collection:Traceback (most recent call last): File "/opt/em7/lib/python3.6/proc_pda.py", line 215, in _init_ File "/opt/em7/lib/python3.6/collect_cdp.py", line 163, 84
silo.cdp_collect.33: Unhandled exception on appliance: 12, process Data Collection: CDP Collection:Traceback (most recent call last): File "/opt/em7/lib/python3.6/proc_pda.py", line 215, in _init_ File "/opt/em7/lib/python3.6/collect_cdp.py", line 163  84
silo.cdp_collect.33: Unhandled exception on appliance: 13, process Data Collection: CDP Collection:Traceback (most recent call last): File "/opt/em7/lib/python3.6/proc_pda.py", line 215, in _init_ File "/opt/em7/lib/python3.6/collect_cdp.py", line 163  84
silo.cdp_collect.33: Unhandled exception on appliance: 10, process Data Collection: CDP Collection:Traceback (most recent call last): File "/opt/em7/lib/python3.6/proc_pda.py", line 215, in _init_ File "/opt/em7/lib/python3.6/collect_cdp.py", line 163  84
silo.subscription_usage_crunch.153: Unhandled exception on appliance: 1, process Enterprise Database: Subscription Usage Crunch:Traceback (most recent call last): File "/opt/em7/backend/process/proc_pda.py", line 231, in _init_ File "/opt/em7/proc/sub 7
silo.dynamic_check.28: Unhandled exception during collection, process Data Collection: Dynamic Refresh:Traceback (most recent call last): File "/tmp/pip-install-lTucNu/sl-collect/sl_collect/shared.py", line 199, in _run_collect_async File "/usr/lib64/py  7
silo.dynamic_check.28: Unhandled exception during collection, process Data Collection: Dynamic Refresh:Traceback (most recent call last): File "/tmp/pip-install-lTucNu/sl-collect/sl_collect/shared.py", line 199, in _run_collect_async File "/usr/lib64/py  5
silo.dynamic_check.28: Unhandled exception during collection, process Data Collection: Dynamic Refresh:Traceback (most recent call last): File "/tmp/pip-install-lTucNu/sl-collect/sl_collect/shared.py", line 199, in _run_collect_async File "/usr/lib64/py  5
silo.dynamic_check.28: Unhandled exception during collection, process Data Collection: Dynamic Refresh:Traceback (most recent call last): File "/tmp/pip-install-lTucNu/sl-collect/sl_collect/shared.py", line 199, in _run_collect_async File "/usr/lib64/py  3
Total   699
```

2024-01-09 EM-62605: Unhandled exception raised during CDP topology collection related to an invalid IP address

ts_guide_to_jira_link

Link: EM-62605: Unhandled exception raised during CDP topology collection related to an invalid IP address CLOSED

From the description:

```
I hit these unhandled exceptions on my newly-built 12.2.1 distributed stack. It looks like an IP address collected from one of the SNMP devices I was discovering was not liked by sl_network, when called by
 CDP collection. We'll want to handle this exception in sl_topology before we ship 12.2.1. See lines 10 and 11 for the exceptions - it appears there may be two places where we need to catch these exceptions
```

```
and log them, but continue.
```